CE 310 — Week 2 Lab: Shape Matters — Reading Distributions

Dataset: Tucson Airport Hourly Weather, 2023 (same file as Week 1) **Topic:** Histograms · Skewness · Quartiles · Percentiles

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Before You Begin

Files needed:

- `Week1_Tucson_Weather_2023.csv` — same file from Week 1
- `Wk02.04_In_Class_Answer_Sheet.xlsx` — download from D2L and enter all answers here

Answer Sheet column map:

Column	What to enter
Col B	(pre-filled question label — do not edit)
Col C	Your Excel formula or method description
Col D	★ Intermediate value (★ questions only — leave blank otherwise)
Col E	**Your numerical answer**
Col F	(pre-filled unit — do not edit)

Your answer cell matches the question label — A1 goes in row A1 of Col C, B3 in row B3.

> **Type your formula as text.** In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, `'=AVERAGE(B2:B8761)`. Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted**; the typed formula is what earns the approach credit.

Excel Quick Reference

> **`Data!` is not optional.** Your tables live on `Work` and your answers on `Answers`, but the readings live on `Data`. Every range you type outside the `Data` sheet needs the sheet name in front of it — `Data!B2:B8761`, not `B2:B8761`. Without it Excel looks on the sheet you are standing on, finds nothing, and hands back 0 or `#DIV/0!`. Most ranges below are written bare; add the prefix yourself. This is the same rule as Week 1.

Goal	Formula / Method
Count values in bin	`=COUNTIFS(Data!B$2:B$8761, ">="&A7, Data!B$2:B$8761, "<"&(A7+10))`
Build histogram chart	From the bin table: select `B6:C16` → Insert → Column → Clustered Column, Gap Width 0%
Skewness coefficient	`=SKEW(B2:B8761)`
First quartile (Q1)	`=QUARTILE(B2:B8761, 1)` or `=PERCENTILE(B2:B8761, 0.25)`
Third quartile (Q3)	`=QUARTILE(B2:B8761, 3)` or `=PERCENTILE(B2:B8761, 0.75)`

Goal	Formula / Method
Nth percentile	`=PERCENTILE(B2:B8761, n/100)` — e.g., 0.90 for 90th percentile
Monthly average	`=AVERAGEIF(E2:E8761, 8, D2:D8761)` — month 8 = August
Count above threshold	`=COUNTIF(B2:B8761, ">100")`

This week you'll build the tools to read a distribution's shape, not just its center.

Setup (~10 min)

You do all of this week's work in **one file: your Answer Sheet.** The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in. Add as many sheets as you like — only `Instructions`, `Answers`, `Work`, `Charts` and `Written` have to keep their names.

1. Open your Answer Sheet and add a sheet named `Data`. Re-import `Week1_Tucson_Weather_2023.csv` onto it, or paste in the data from your Week 1 file.
2. Confirm your column layout:
   - Column A: `DateTime`
   - Column B: `Temp_F`
   - Column C: `Wind_Speed_mph`
   - Column D: `Relative_Humidity_pct`
   - Column E: `Month` — the helper column you built in Week 1 with `=MONTH(A2)`
   - If column E is empty, build it now: type `Month` in E1, `=MONTH(A2)` in E2, and fill down to row 8761. Every formula this week that filters by month reads column E, so that is where it has to live.
3. **Quick calculation — Variance:** Before starting Section A, compute the temperature variance in `Work!B32`, in the block headed *Variance check*:

   ```
   =VAR(Data!B2:B8761)
   ```

   You should get approximately **320.8 °F²**. Note the relationship: VAR = STDEV². Your Week 1 A5 computed STDEV ≈ 17.91°F, and 17.91² = 320.8. Variance and standard deviation describe the same spread — variance has squared units (°F²), which are hard to interpret directly, so engineers almost always report standard deviation instead. Week 2 introduces skewness, which captures something variance and stdev cannot: the asymmetry of the distribution.

Section A — Histogram Basics (~25 min)

Section A builds the two instruments that reveal a distribution's shape: the histogram, which shows it, and the skewness coefficient, which measures it.

Bin counts → whole numbers. SKEW coefficients → 2 decimal places. CV (A9) → 4 decimal places.

Counting Histogram Bins

`COUNTIFS` counts rows meeting two conditions at once — a lower bound and an upper bound:

```
=COUNTIFS(Data!B2:B8761, ">="&60, Data!B2:B8761, "<"&70)
```

The lower boundary is >= (inclusive), the upper is < (exclusive), so each hour lands in exactly one bin.

Build the table on the `Work` sheet of your Answer Sheet, in the block headed *Questions A1-A3 + Chart 1*. Column A already holds each bin's lower edge as a number and column B its label; you fill column C. Type the formula once in `C7` and double-click the fill handle to carry it to `C16` — that is what the numeric column A is for, and it is exactly what the Excel Tutorial walked through. Chart 1 plots columns B and C, and A1–A3 read three rows of it.

To chart it: select the bin labels and counts, then **Insert → Column → Clustered Column**, and set **Gap Width** to 0% so the bars touch. The Excel Tutorial shows both steps, and which neighbouring button to avoid.

A1 (2 pts) — Temperature Histogram: 60–70°F Bin

Using the COUNTIFS method above, find the count of hours where temperature was in the **60–70°F** range (≥ 60°F and < 70°F).

A2 (2 pts) — Temperature Histogram: 80–90°F Bin

Find the count of hours in the **80–90°F** range.

This is the **modal bin** — the most common temperature range in Tucson's 2023 data.

A3 (2 pts) — Temperature Histogram: 70–80°F Bin

Find the count of hours in the **70–80°F** bin.

> Your bin convention is a choice, and it changes the answer. `COUNTIFS` as written counts [70, 80) — at or above 70, below 80. Tools that count (70, 80] instead put the boundary hours in the neighbouring bin, which shifts this count by more than 100 hours on this dataset. State the convention you used, and keep it the same for every bin.

A4 (2 pts) — Temperature Skewness

Calculate the skewness coefficient for the temperature column using `SKEW`.

> Near zero, so the shape is close to symmetric. The modal bin (A2) sits well above the mean of ~72°F, which usually signals left skew — here it is too slight to register.

A5 (2 pts) — Wind Speed Histogram: 0–5 mph Bin

Build a second frequency table for **wind speed** in the `Work` sheet block headed *Questions A5-A6 — Wind speed histogram bins*. Same pattern, filled from `C21` down to `C27`, with two changes you have to make yourself:

- the data column moves from `Data!B` (temperature) to **`Data!C`** (wind speed)
- these bins are **5 mph wide, not 10**, so the upper bound is `"<"&(A21+5)`

Paste the temperature formula unchanged and the 0–5 bin comes back around 6,700 instead of roughly 3,000 — it has counted a 10 mph band and read the wrong column.

Find the count of hours in the **0–5 mph** bin.

A6 (2 pts) — Wind Speed Histogram: 5–10 mph Bin

From your wind speed table, find the count of hours in the **5–10 mph** bin.

> Larger than the 0–5 bin? That places the typical wind speed, not the calm hours, at the peak.

A7 (2 pts) — Wind Speed Skewness

Use `SKEW` on the wind speed column.

> Positive means right-skewed: most hours calm, a long tail of high winds. Compare its size to A4.

A8 (2 pts) — Humidity Skewness

Use `SKEW` on the relative humidity column.

A9 (2 pts) — Coefficient of Variation (CV)

CV = σ / μ, using `STDEV` and `AVERAGE` on the temperature column. Being dimensionless, it compares variability across quantities with different units.

> Above 0.20, designing for the average misses the tails badly. Tucson lands above 0.24; San Francisco sits below 0.10.

Section B — In-Depth Analysis (~40 min)

Section B asks what the shape costs you. Percentiles take over from the average as the design value, and each question contrasts what the mean would have told you with what the distribution actually says.

Report temperatures in °F, speeds in mph, percentages in %.

B1 (4 pts) — Temperature Q1 (Verify with PERCENTILE)

Use `PERCENTILE` to confirm the Q1 you found in Week 1 (A10 ≈ 58°F). Its second argument is a decimal between 0 and 1.

> Same answer as `QUARTILE(range, 1)` — Q1 *is* the 25th percentile.

B2 (4 pts) — Temperature Q3 (Verify with PERCENTILE)

Verify your Week 1 Q3 (answer A11 ≈ 86°F) using `PERCENTILE` at the 75th percentile.

B3 ★ (5 pts) — Temperature IQR

`IQR = Q3 − Q1`

Col D: Q1 (from B1) · **Col E:** IQR = Q3 − Q1

> `QUARTILE` only reaches Q1, Q2 and Q3. B4 and B5 need the 90th and 95th, which is why this week introduces `PERCENTILE`.

B4 (4 pts) — Temperature 90th Percentile

Use `PERCENTILE` on the temperature column at the 90th percentile.

B5 ★ (5 pts) — Temperature 95th Percentile

Use `PERCENTILE` on the temperature column at the 95th percentile.

Col D: 90th percentile (from B4) · **Col E:** 95th percentile

B6 (4 pts) — Hours Above the 95th Percentile

How many hours had a temperature **strictly above** your B5 value? Use `COUNTIF` with your B5 *value* as the threshold — the number you computed, not the label `B5`. Join it to the operator with `&`, the same way the bin formula does; written as `">B5"` in quotes it counts nothing.

> This should land near your Week 1 A7 (hours > 100°F) and near 5% of 8,760 ≈ 438. A small gap from 438 is PERCENTILE interpolating between discrete readings, not an error.

> **Engineering Context:** ASCE 7-22 requires structural wind design based on a 50-year return period wind speed — a threshold far more extreme than the 95th percentile of a single year.

B7 (4 pts) — Wind Speed Q1

Use `QUARTILE` or `PERCENTILE` on the wind speed column.

B8 (4 pts) — Wind Speed Q3

Use `QUARTILE` or `PERCENTILE` on the wind speed column.

B9 (4 pts) — Wind Speed IQR

`IQR = Q3 − Q1`

> Compare with the temperature IQR (B3): which has more spread through its middle 50%, and does that match their SKEW values?

B10 (4 pts) — Wind Speed 95th Percentile

Use `PERCENTILE` on the wind speed column at the 95th percentile.

B11 ★ (5 pts) — Wind Pressure at 95th Percentile

Using the simplified ASCE wind pressure formula from Week 1 (Cd = 1.3):

`P = 0.00256 × 1.3 × V²`

where V is your B10 value.

Col D: V^2 (B10 squared) · **Col E:** P in psf

At the annual average wind speed of ~7.24 mph, `P_avg ≈ 0.174 psf`. The ratio between the two is why average-based wind design fails.

> **ArcE Framing:** ASHRAE 62.1 requires HVAC latent load sizing to outdoor design humidity — the 90th percentile humidity, not the annual average.

B12 (4 pts) — Humidity 90th Percentile

Use `PERCENTILE` on the relative humidity column at the 90th percentile.

B13 (4 pts) — August Average Humidity (Peak Monsoon)

August is Tucson's peak monsoon month. Calculate the average relative humidity for August only, using `AVERAGEIF` with the Month helper column (column E, month = 8).

B14 ★ (5 pts) — Probability Bridge: P(80°F ≤ T < 90°F)

Estimate the probability that a random hour in 2023 fell between 80°F and 90°F:

`P = (count of hours in 80–90°F bin) / (total hours) × 100`

Col D: count of 80–90°F hours (your A2) · **Col E:** probability as a percentage to two decimals — write `7.65`, not `0.0765` and not `7.65%`

Required Charts (15 pts)

Both charts belong on the **`Charts` sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart** -> **Object in** -> `Charts`, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — Temperature Histogram

Chart the bin table you built on the `Work` sheet: select **`B6:C16`** (labels and counts, headers included — not column A), then **Insert → Column → Clustered Column**, and set **Gap Width to 0%** so the bars touch.

Do not reach for Insert → Statistical Charts → Histogram here. That command bins raw values itself, so it plots `Data!B2:B8761` and chooses its own bin edges — your table is not in the picture. Chart 1 is scored by looking at the cells the chart plots, and those cells have to be your ten counts.

- Title: `Hourly Temperature Distribution — Tucson, AZ 2023`
- Axes: `Temperature (°F)` on x, `Number of Hours` on y
- Add a text box marking `Mean ≈ 72°F` near the 70–80°F bar

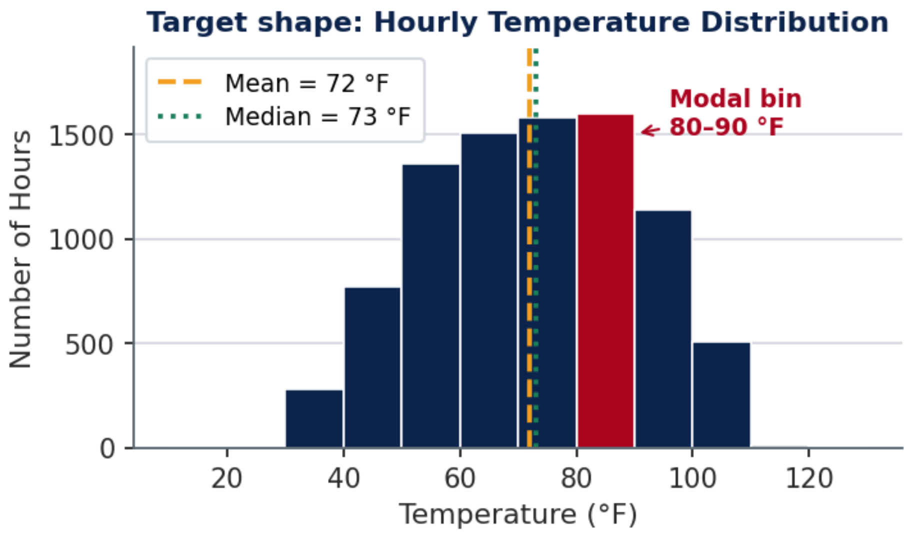

Figure 1: Target shape for Chart 1

Does this histogram show one peak or two? A second distinct peak would signal mixed subpopulations requiring separate analysis — a key EDA diagnostic.

Chart 2 (8 pts) — Temperature Box-and-Whisker Chart

Build a box plot of `Temp_F` (B2:B8761) with **Insert → Statistical Charts → Box and Whisker**, straight from the raw column.

- Title: `Temperature Box Plot — Tucson, AZ 2023`
- Y-axis: `Temperature (°F)`
- Add a text box with `IQR = [your B3 value] °F`

Does the median sit near the center of the box? Do any outlier dots appear outside the whiskers?

Memo Deliverable (7 pts)

Type your memo directly in the **Written sheet** of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed. Use the template below.

```
To:      CE 310 Instructor
From:    [Your Name], [Your Major]
Date:    [Date]
```

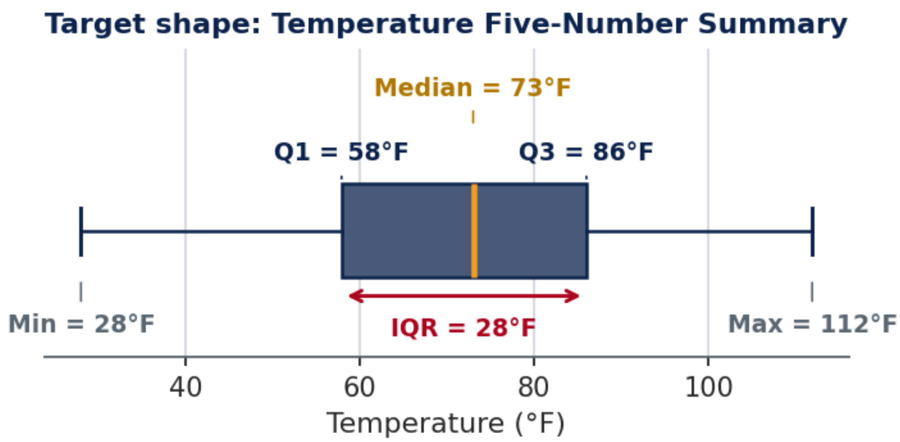

Figure 2: Target shape for Chart 2

`Subject: Week 2 Lab — Distribution Shape Analysis, Tucson 2023 Weather Data`

Your three paragraphs each map to an EDA goal: Paragraph 1 → Describe (center, spread, shape); Paragraph 2 → Discover (cross-variable patterns); Paragraph 3 → Check (design conditions).

Paragraph 1 (3 pts) — Temperature Distribution Shape: Describe the shape of the temperature distribution using your histogram (Chart 1) and SKEW coefficient (A4). Reference your box plot (Chart 2) — state where the median falls relative to Q1 and Q3, and what the box width tells you about the spread of typical annual temperatures. **Give your IQR (B3) as a number**; that value is scored on its own, so "a wide box" does not earn it. Explain why the shape looks like a plateau rather than a bell curve.

Paragraph 2 (2 pts) — Comparing Variables: Compare the temperature distribution (A4 SKEW) to the wind speed distribution (A7 SKEW) and humidity distribution (A8 SKEW). **Cite at least two of the three coefficients as numbers**, say which variable is most skewed, and connect that to why its design value has to come from a high percentile rather than the mean. Then give the physical reason: why is wind speed right-skewed while temperature is nearly symmetric?

Paragraph 3 (2 pts) — Latent cooling load and humidity design conditions:

Using your B12 (humidity 90th percentile) and B13 (August average humidity), explain the latent cooling load challenge for a Tucson office building. Compare these values to the annual average humidity (32.67%). Describe what happens to indoor air quality and occupant comfort if the HVAC system is sized for the annual average instead of the 90th percentile condition. Reference **ASHRAE 62.1** and the concept of design conditions.

Submission: Save as `CE310_W2_<NetID>.xlsx` (e.g., `CE310_W2_abc123.xlsx`) and upload to the Week 2 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the `Answers` sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
- ☐ File saved as `.xlsx` and named `CE310_W2_<NetID>.xlsx`
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
- ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
- ☐ ★ questions also have their intermediate value in **Col D**
- ☐ Both charts sitting in their boxes on the `Charts` sheet, as live charts and not pictures, with titles and axis labels
- ☐ Memo written in the `Written` sheet
- ☐ Uploaded to D2L by **9:00 AM next Wednesday**

Grading Summary

Component	Points
Section A — A1–A9	18
Section B — B1–B14	60
Chart 1 — Temperature histogram	7
Chart 2 — Temperature box-and-whisker chart	8
Memo (3 paragraphs)	7
Total	**100**
